

TaskFlow - Task Management Assessment

TaskFlow is a full-stack task management web application built with React, Node.js, Express, JWT authentication, and MongoDB.

Tech Stack

- Frontend: React 19, TypeScript, Vite, Context API, CSS
- Backend: Node.js, Express.js, MongoDB, Mongoose
- Auth: JWT with protected routes and bcrypt password hashing
- Extras: Search, pagination, dark mode, admin-only route, Swagger API docs, Docker Compose, Jest unit test

Folder Structure

```
Task-Management-Assessment/  
  Backend/  
    src/  
      config/           MongoDB connection  
      controllers/      Route handlers  
      docs/             Swagger document  
      middleware/       Auth, validation, error handling  
      models/           Mongoose schemas  
      routes/           Express routers  
      validators/       Zod request schemas and tests  
  Frontend/  
    src/  
      components/       Auth, dashboard, task form/card  
      context/           Auth context  
      services/          API client  
      types.ts           Shared frontend types
```

Local Setup

1. Install dependencies

```
npm run install:all
```

2. Configure backend environment

Create Backend/.env :

```
PORT=5000  
MONGODB_URI=MONGODB_URI  
JWT_SECRET=JWT_SECRET  
JWT_EXPIRES_IN=7d  
CLIENT_URL=CLIENT_URL
```

3. Configure frontend environment

Create Frontend/.env :

```
VITE_API_URL=http://localhost:5000/api
```

4. Start MongoDB

Use a local MongoDB instance or run:

```
docker compose up mongo
```

5. Start the app

In two terminals:

```
npm run dev:backend  
npm run dev:frontend
```

Frontend: <http://localhost:5173>

Backend health check: <http://localhost:5000/api/health>

Swagger docs: <http://localhost:5000/api/docs>

Docker

To run MongoDB and the backend API:

```
docker compose up --build
```

Run the frontend locally with `npm run dev:frontend`.

API Details

Base URL: `http://localhost:5000/api`

Method	Endpoint	Access	Description
POST	<code>/auth/signup</code>	Public	Register a user
POST	<code>/auth/login</code>	Public	Login and receive JWT
GET	<code>/auth/me</code>	Private	Get current user
GET	<code>/tasks</code>	Private	List own tasks with <code>status</code> , <code>search</code> , <code>page</code> , <code>limit</code>
POST	<code>/tasks</code>	Private	Create a task
PATCH	<code>/tasks/:id</code>	Private	Update own task
DELETE	<code>/tasks/:id</code>	Private	Delete own task
GET	<code>/tasks/admin/all</code>	Admin	List all users' tasks

Authenticated requests require:

```
Authorization: Bearer <jwt>
```

Testing

```
npm test
```

Assumptions

- Users can only manage their own tasks.
- Newly registered users receive the `user` role by default.
- Admin users can be promoted directly in MongoDB by setting `role` to `admin`.
- Pagination defaults to 8 tasks per page on the frontend.